

Solution Requirements

Date	05-10-2026
Team ID	6ab2256ffc67a5bc022256a7
Project Name	Script_ Controlled ACL _ Restrict Record Access Based on Field Value
Maximum Marks	4 Marks

Define Functional and Non-Functional Requirements:

Create a comprehensive list of requirements to define exactly what your solution will do and how well it will perform.

- **Functional Requirements (FR)** define specific behaviors, functions, or features of the system (What the system should do).
- **Non-Functional Requirements (NFR)** specify the criteria that judge the operation of a system, rather than specific behaviors (How the system should be).

Step 1: Functional Requirements (FR)

S.No	Requirement Category	Requirement Description	Priority (High/Medium/Low)
1	Authentication	Authenticate users via standard ServiceNow SSO / OAuth2 credentials before evaluating ACL script conditions.	High
2	Authorization levels	Define granular roles (e.g., admin, record_manager, end_user) and grant override privileges to security administrators.	High
3	External interfaces	Interface with ServiceNow REST APIs and Scripted REST Endpoints to enforce ACL policies on external integrations.	Medium
4	Transactions processing	Dynamically intercept read/write transactions and evaluate target field values (e.g., state == 'Closed' or 'Archived') in real time.	High
5	Reporting	Provide administrative logs and reports summarizing ACL evaluation passes, rejections, and access violation attempts.	Medium
6	Business rules, etc.	Execute server-side JavaScript within ACL rules to restrict access dynamically whenever specific record state thresholds are met.	High
7	Compliance to laws or regulations	Ensure data privacy compliance (GDPR, HIPAA) by preventing unauthorized access or editing of sensitive archived records.	High
8	Other	Display custom user error notifications when record editing is blocked due to field-based script restrictions.	Low

Step 2: Non-Functional Requirements (NFR)

S.No	NFR Category	Requirement Description	Target Metric / Acceptance Criteria
1	Performance & Speed	Scripted ACL execution overhead must be minimal to ensure fast form render and response times.	ACL script evaluation completes in < 50ms per record request.
2	Scalability	Maintain consistent ACL rule performance under high transaction volumes and concurrent active sessions.	Supports up to 5,000 concurrent record evaluation requests.
3	Security & Data Privacy	Prevent unauthorized data exposure and script execution bypasses through secure server-side evaluation.	Zero unauthorized read/write access incidents verified via security audits.
4	Reliability & Availability	Ensure continuous operational availability of security rules and ACL script logic across instance nodes.	99.9% availability during active system operations.
5	Usability & Accessibility	Provide clear visual cues (e.g., read-only lock icons) on restricted fields for end users.	100% adherence to standard ServiceNow UI/UX guidelines.

6	Other	Maintain modular, reusable, and well-commented JavaScript ACL code for ease of future updates.	Passes standard code review and static security analysis.
---	--------------	--	---